Project: RouteRanker – AI-Powered Public Transit Route Optimization

Project Statement:

Develop an AI system that analyzes public transit data to optimize inefficient bus routes across Indian cities using machine learning. The system will identify underperforming or overcrowded routes, recommend changes (e.g., merging, frequency adjustment), and provide insights through a Streamlit-based interface.

Outcomes:

Clear identification of route inefficiencies.
Strategic optimization recommendations.
User-friendly interface for planners to view and act on model insights.

Dataset Sources:

Source: Lepton Software – GTFS for Indian Transit(https://datarade.ai/data-products/india-transit-data-railways-metro-and-public-buses-routes-lepton-software)
Coverage: 23 Indian cities (Delhi, Bengaluru, Mumbai, Hyderabad, Pune, etc.)
Data Format: General Transit Feed Specification (GTFS)
Files Used:

- `routes.txt`: List of public transit routes
- `trips.txt`: Trip-level data linked to routes
- `stop_times.txt`: Stop sequence and timing for each trip
- `stops.txt`: Geographical stop locations
- `calendar.txt`: Operational schedule
- Optional: `frequencies.txt`, `shapes.txt`

Modules:

1. Data Collection.
2. Feature Engineering.
3. Exploratory Data Analysis (EDA).
4. Data Preprocessing.
5. Model Selection & Training.
6. Inference & Route Suggestion.
7. Frontend Development (Streamlit App).

Milestones:

Milestone 1: Data Collection and Feature Engineering(Weeks 1–2)
Week 1:
Collect GTFS datasets from Lepton Software for 23 major Indian cities.
Extract and load essential files: routes.txt, trips.txt, stop_times.txt, stops.txt, calendar.txt.
Perform initial data inspection and understand dataset structure.
Week 2:
Clean data by handling missing values and correcting erroneous entries.
Engineer time-based features such as hour of day, weekday vs weekend.
Aggregate stop usage data to calculate total passengers per route per hour.
Construct basic route profiles including route length and trip frequency.

Milestone 2: E.D.A and Data Processing (Weeks 3–4)
Week 3:
Conduct exploratory data analysis (EDA) to identify high-traffic time windows and route usage patterns.
Visualize data with route density heatmaps and hourly ridership curves.
Detect overlapping or underutilized routes through visual insights.
Week 4:
Preprocess data by normalizing/scaling features and encoding categorical variables like route IDs.
Prepare datasets formatted for clustering and classification model training.
Split data into training and validation sets.

Milestone 3: Model Selection and training and Inference (Weeks 5–6)
Week 5:
Train clustering models (KMeans, DBSCAN) to group routes with similar usage profiles.
Train classification models (XGBoost, Random Forest) to predict route performance (e.g., balanced vs overcrowded).
Evaluate model performance using appropriate metrics and validate results.
Week 6:
Run inference on new/unseen data using trained models to identify inefficient routes.
Generate route optimization suggestions for merging, splitting, or adjusting trip frequency.
Compare current metrics with optimized outcomes to measure impact.

Milestone 4: Visualization and Final Report (Weeks 7–8)

Week 7:

Develop a Streamlit dashboard to upload GTFS files, visualize route clusters, and show model predictions.

Implement interactive visualizations for route optimization scenarios and metrics comparison.

Prepare final technical documentation and stakeholder presentation summarizing the project outcomes.

Week 8:

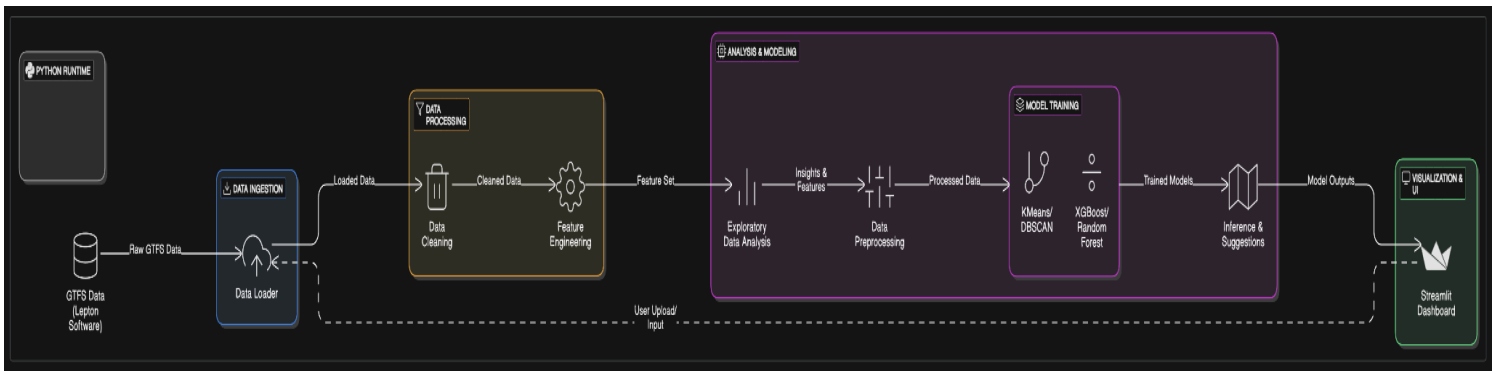
Document system architecture and model outcomes.

Deliver stakeholder presentations and deployment recommendations.

Workflow:

1. Data Collection.
2. Data Cleaning & Feature Engineering.
3. Exploratory Data Analysis (EDA).
4. Data Preprocessing.
5. Model Development.
6. Model Inference & Optimization.
7. Dashboard & Visualization.
8. Reporting & Deployment.

Architecture Diagram:



Tools & Tech Stack:

Language:

Python

Libraries:
Pandas, NumPy, sci-kit learn, XGBoost, Matplotlib, Seaborn, Plotly, Geopandas, Folium.

ML Models:
KMeans, DBSCAN, XGBoost, Random Forest

Visualization & UI:
Streamlit or Dash.

Expected Outcomes
Clear identification of route inefficiencies.
Strategic optimization recommendations.
User-friendly interface for planners to view and act on model insights

Evaluation Criteria:

Milestone Completion.
Successful delivery of each stage with corresponding modules tested and validated.

Optimization Effectiveness
Comparison between current vs simulated route loads.

System Integration and Visualization
Functional dashboard.

Presentation and Documentation
Technical report, methodology writeup, and stakeholder-ready presentation.